

Homework 01

Given the following matrices representing the orientation of three different frames in the space:

$$R_1 = \begin{bmatrix} -0.9218 & 0.0078 & 0.3875 \\ 0.0078 & -0.9992 & 0.0389 \\ 0.3875 & 0.0389 & 0.9211 \end{bmatrix}$$

$$R_2 = \begin{bmatrix} -0.9641 & 0.1007 & 0.2458 \\ 0.1007 & -0.7180 & 0.6888 \\ 0.2458 & 0.6888 & 0.6820 \end{bmatrix}$$

$$R_3 = \begin{bmatrix} 0.9363 & -0.2751 & 0.2184 \\ 0.2896 & 0.9564 & -0.0370 \\ -0.1987 & 0.0978 & 0.9752 \end{bmatrix}$$

$$R_4 = \begin{bmatrix} 0 & -0.6428 & 0.7660 \\ 0 & 0.7660 & -0.6428 \\ 1 & 0 & 0 \end{bmatrix}$$

- Verify that R_1, R_2, R_3, R_4 , are rotation matrices;
- For each matrix, determine the corresponding axis-angle and unit-quaternion representations;
- For each matrix, determine the minimal representation of orientation according to ZYZ and RPY angles.

Homework 02

- Demonstrate the relationship which expresses $R_r(\theta)$ in slide #21 of "02 Kinematics - Part 1"
- Demonstrate the relationship which expresses $R(\eta, \epsilon)$ in slide #33 of "02 Kinematics - Part 1"